

Robust health-score based survival prediction for a neonatal mouse model of polymicrobial sepsis

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Order of Authors:	Byron Brook Danny Harbeson Nelly Amenyogbe Rym Ben-Othman Tobias R. Kollmann Radhouane Aniba
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Tuesday, October 23, 2018

We would like to submit to you for publication in *PLoS ONE* our manuscript entitled '*Robust health-score based survival prediction for a neonatal mouse model of polymicrobial sepsis*' by Byron Brook, Danny Harbeson, Nelly Amenyogbe, Rym Ben-Othman, Tobias R. Kollmann, and Radhouane Aniba as a **research article**. This work presents an objective, statistically well-supported and easily applied tool to **predict outcome** prior to reaching the **humane endpoint** in murine neonatal sepsis. Implementation of this tool will **enable assessing** cause-effect chains prior to reaching the experimental endpoint, and with that lead not only to a better understanding of the **relevant mechanisms** causing death in **sepsis**, but also the rapid **testing** of interventions. **Similar 'predictors'** had been **established** and are in wide use in **adult mouse** research projects focused on sepsis; however, these adult models **cannot** simply be applied to **newborns**. Thus, models predicting outcome had not been available to scientists working with newborn mice. Of note, this approach allows determination of the humane endpoint not only for newborn sepsis but for any **newborn mouse** research project, thus will be extremely useful for veterinarians and ethics committees charged with ensuring the wellbeing of newborn mice, as implementation would reduce needles suffering of the animal.

This manuscript **has not been published** previously and is being submitted only to *PLOS ONE*. All authors are responsible for reported research and have participated in the **concept** and **design**, **analysis** and **interpretation** of data, as well as **drafting** of this manuscript.

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If there is anything else we can do to help with this manuscript submission, please do let us know. My contacts as the corresponding author for all further communication is listed below.

Yours sincerely,

Danny Harbeson

PhD Candidate, Experimental Medicine
Department of Medicine
University of British Columbia
BC Children's Hospital Research Institute, A5-147
950 W 28th Ave, Vancouver, V5Z 4H4 Canada
604-340-5627; dannyharbeson@gmail.com

Robust health-score based survival prediction for a neonatal mouse model of polymicrobial sepsis

Byron Brook^{1¶}, Danny Harbeson^{1¶*}, Nelly Amenyogbe¹, Rym Ben-Othman², Tobias R. Kollmann^{2&},
Radhouane Aniba^{3,4&}

¹ Department of Experimental Medicine, University of British Columbia, Vancouver, BC, Canada

² Department of Pediatrics, Division of Infections Diseases, University of British Columbia, Vancouver,
BC, Canada

³ Department of Molecular Oncology, BC Cancer Agency, Vancouver, BC, Canada.

⁴ Department of Pathology and Laboratory Medicine, University of British Columbia, Vancouver, BC,
Canada

* Corresponding author

E-mail: dannyharbeson@gmail.com (DH)

[¶]These authors contributed equally to this work

[&]These authors also contributed equally to this work

Abstract

Infectious disease and sepsis represent a serious problem in early life. Much of the associated morbidity and mortality is presumed to relate to fundamental differences between neonatal and adult immunity. Mechanistic insight into the way newborns' immune systems handle infectious threats is lacking; as a result, there has only been limited success in providing effective immunomodulatory interventions to reduce mortality. Given the complexity of the host-pathogen interactions, neonatal mouse models can offer potential avenues providing valuable data. However, the small size of neonatal mice hampers the ability to collect biological samples without sacrificing the animals. Further, the lack of a standardized metric to quantify mouse health increases reliance on correlative biomarkers without a known relationship to 'clinical' outcome. To address this bottleneck, we developed a system that allows assessment neonatal mouse health in a readily standardized and quantifiable manner. The resulting health scores require no special equipment or sample collection and can be assigned in less than 20 seconds. Importantly, the health scores are highly predictive of survival. A classifier constructed from the righting reflex and mobility revealed a positive relationship between bacterial load and survival, demonstrating how this scoring system can be used to bridge the gap between assumed relevance of biomarkers and the clinical outcome of interest. Adoption of this scoring system will not only provide a robust metric to assess health of newborn mice but allow for objective, prospective studies of infectious disease and possible interventions in early life.

Introduction

An estimated 3.0 million cases of newborn infection every year result in nearly 350,000 deaths, making infection one of the most lethal threats in early life [1]. It is widely known that the neonatal immune system is fundamentally distinct from that of adults, indicating that data generated from adult animal models should not be assumed to apply to newborns [2–5]. While neonatal mouse models offer several practical benefits (availability, existing infrastructure, well-understood biology, etc.), working with neonatal mice presents some unique challenges which have not yet been effectively addressed. In particular, researchers are limited by the size of the mice; near day of life (DOL) 7 (considered to be immunologically closest to human newborns) [6] mice are still far too small to collect meaningful volumes of blood without sacrificing experimental animals. In adult mice, about 7 to 8 μL of blood can ethically be collected per gram of mouse bodyweight without sacrifice or fluid replacement – this translates to a maximum single collectible blood volume from neonatal mice of about 20 μL [7]. Serial sampling (every 24 hours) would only allow for collection 5 μL of blood or less in each draw [8]. This limitation (in addition to the practical difficulties of drawing blood from such small animals) leads to sample collection at time of sacrifice, which fundamentally separates biological findings from true ‘clinical’ outcomes (i.e. survival). This inability to serially sample neonatal mice without sacrifice also forces researchers to rely heavily on biomarkers as a stand-in for outcome when exploring a potential treatment of interest, assuming causality between e.g. pathogen load or inflammatory cytokine production and mortality.

Health scores for sepsis are widely used in human clinical settings as they are critical for capturing disease progression and can inform potential interventions [9–13]. The ability to assess disease severity in a quantitative way is also extremely useful in animal models as it can help investigators avoid an

overreliance on assumptions about the relevance of a given biomarker. For example, one group developed a series of **symptom scores** for adult mice in a similar model of sepsis (fecal suspension injected IP) which relied on, among other things: **activity**, **response to stimulus**, **eyes** (open or shut), **level** of **consciousness**, and **appearance** [14]. This study illustrated that when **grouping adult mice** by these **health scores**, **TNF α** and **IL-1 β** levels did **not correlate with outcome** despite elevated levels in comparison to unchallenged controls – these cytokines went up following challenge but did appear different in survivors and non-survivors. This demonstrates the capacity for **health scores** to **narrow** the focus of investigators **onto biomarkers** which are **functionally relevant** to a **clinical outcome** of interest – a capacity even more critical when working in a neonatal population from which even **blood samples** are **extremely difficult** to **collect** without harming them. While the **quantitative** observations used in this model may be strong markers for adult animals, **none** of these characteristics are **readily applicable** to a **neonatal population** (i.e. **no fur** to be ruffled, **eyes** always closed, minimal **basal activity** levels, etc.).

no
?

We developed a **data-driven** and **simple scoring system** that is directly **related to outcome** (humane endpoint) in neonatal mice. The addition of our **health scores** to neonatal mouse experiments provides:

(a) **improved tracking** of **disease progression** which can inform time of sacrifice for sample collection and

(b) a **quantitative method** to **prospectively differentiate survivors from non-survivors**. We here validated this **six-point health scoring system** (0-5) in a **polymicrobial model** of neonatal sepsis. This model confirms that **clinical outcome** is **highly correlated** with **bacterial load**. It takes less than 20 seconds to determine these composite **health scores** per animal. The prospective **validity** of our **health score** in assigning outcome provides the much-needed **metric** to begin **assessing causality** of **biomarkers**, which in turn will allow assignment of mechanistic cause-effect relationships. This approach will not only

decrease undue suffering of the animals, but also increases the likelihood of identifying effective immunomodulatory interventions for infectious disease and sepsis.

Materials and Methods

Cecal slurry model of neonatal sepsis

Polymicrobial sepsis was induced using a modified version of the model first described by Wynn *et al.* in 2007 [15]. Briefly, cecal material was extruded from adult male mouse ceca (aged 6-10 weeks), diluted in dextrose 5% water (D5W) to 160 mg slurry / mL D5W, filtered through a 70 µm sterile strainer, pooled, aliquoted, and frozen at -80 °C. Aliquots were thawed at room temperature, diluted further in D5W to the desired weight-adjusted dose, and kept on ice prior to intraperitoneal (IP) injection. The challenge dose was titrated to achieve a lethal dose equal to 50% (LD50) per litter. Neonatal mice were injected IP with cecal slurry on DOL 7 or 8 to induce polymicrobial sepsis. Litters were monitored for morbidity (i.e. humane endpoint) up to 96 hours post challenge (HPC). A separate cohort of mice were sacrificed at approximately 24 HPC with bacterial load in liver, lungs, spleen, and blood calculated by serially diluting the organ homogenates in PBS and drop-plating on 5% sheep's blood agar and counting colony forming units (CFU) after 24 hours of incubation at 37 °C.

Mice

Male and female C57BL/6 mice aged 6 weeks (Jackson Labs, catalogue #664) were purchased from The Jackson Laboratory and used for breeding purposes; all mice used in this study were first or second-generation progeny of Jackson mice bred in-house. This study was carried out in strict accordance with animal ethics guidelines outlined by University of British Columbia and was approved by the Animal Care Committee (protocol numbers A17-0110 and A14-0261). All efforts were taken to minimize animal suffering. Mice were anesthetized with isoflurane prior to euthanasia through CO2 exposure, confirmed

by decapitation. Mice analyzed for the present study were principally untreated controls from other experiments not shown here, hence the large sample size (404 mice total). Humane endpoint was defined as the lack of an attempt to right themselves when placed on their back on both sides (“failure to right; non-mobile”, further explained in main text). Monitoring commenced 12 HPC (i.e. prior to the onset of expected mortality) and continued 2-3 times daily through to experimental endpoint. All mice used were either 7 or 8 days old and weighed an average of 3.7 g (2.25 g - 6.06 g).

Classifier

We built a machine learning model for a binary classification problem, assigning mice to either “survivor” or “non-survivor” categories. In order to identify the best model, we tested our dataset in 6 different algorithms, including baseline learning algorithms (Logistic Regression, K Nearest Neighbors and Decision Tree) as well as ensemble learning algorithms (Random Forest, Gradient Boosting and XGBoost). Various approaches were tested for feature selection, but the final classifier relied on Pearson correlation coefficients. Feature selection and classification was performed on components of the scores (righting reflex and mobility) rather than directly on the scores themselves in order to independently quantify the significance of each metric, in comparison to other commonly collected biometric data (i.e. weight). Gradient Boosted machine learning was selected as the strongest classification algorithm based on accuracy and AUC (Table S1) and was applied to an external dataset for further validation. For a more detailed description of classifier construction, see the supplementary material.

Results

Creating health scores

At all monitoring time points, righting was associated with survival and failure to right (FTR) was associated with non-survival (Fig 1A). As 24 HPC was as far into disease progression as one could wait to sacrifice mice without incurring heavy survivor bias (Fig 1b), this timepoint became the time of sacrifice for the alternative cohort of mice assessed for bacterial load. It was therefore critical to be able to distinguish survivors and non-survivors prior to sacrifice at no later than 24 HPC. Mice which failed to right at 24 HPC significantly more likely to die than those which successfully righted (log-rank test, $p < 0.001$) and had a statistically significant, 100-fold higher bacterial load across all compartments (independent two-group t-tests with Bonferroni correction, $p < 0.001$) (Fig 1c). Righting reflex alone was strongly correlated with survival and bacterial load, especially after 24 HPC. We also captured three different levels of mobility: non-mobile, lethargic, and active (Table 1). One of the strengths of a scoring system should be the ability to provide a metric predictive of future outcome. To assess the predictive value of this righting / mobility combination metric, we assigned numerical scores from 0 (fail to right non-mobile, humane endpoint) to 5 (rights mobile, perfectly healthy) (Table 1). Unless otherwise indicated, the reported score is the lower of the two recorded for each mouse.

other damage?

Figure 1. Righting reflex at 24 HPC is an excellent discriminator of survival and bacterial load.

Righting reflex of neonatal mice (DOL 7-8) challenged IP with cecal slurry at an LD50. (A) Ability to right at different hours post challenge (HPC) in survivors and non-survivors. (B) Survival curve of mice split by reflex at 24 HPC. Mice which were able to right themselves at 24 hours were significantly more likely to survive the cecal slurry challenge than mice which failed to right (log-rank test, $p < 0.001$). (C) Bacterial load split by righting reflex at time of sacrifice, 24 HPC. Mice which were able to right had a significantly lower bacterial load in the whole blood and all measured organ tissues (independent two-group t-tests with Bonferroni correction, $p < 0.001$).

Table 1. Assigning numerical health scores to quantitative observations in neonatal mice

Numerical Score	Righting Reflex*	Mobility	Description
5	Rights	Mobile	- Multiple steps in a row, responsive to tail tap - Alert, may be shaky but moving with energy - Might fall over while walking
4	Rights	Lethargic	- Takes at least one step, but stops between steps - Steps are slower, may be very shaky, may fall over
3	Rights	Nonmobile	- Stationary after righting - May take a half-step after tail tap but largely or entirely unresponsive
2	Fails to right	Mobile	Vigorous hip movement, swinging side to side at an angle >90° from horizontal at least once during the observation period (4 seconds)
1	Fails to right	Lethargic	- Slower hip movement, <90° angle - Halting attempts to right over the observation period, may be some pauses but always starting again
0	Fails to right	Nonmobile	- Zero or extremely weak hip movement - Gives up attempting to right or never starts, may be uncontrollably shaky - limbs may bend at the knee and extend but hips will not rock side to side

*Righting reflex measured by placing mice gently on their back and observing their ability to right within

4 seconds. All scores were taken in duplicate with the lower of the two reported here.

Health scores are strongly associated with survival and bacterial load

Combining mobility with righting reflex allowed us to generate six categories which were assigned to numerical scores for ease of recording and visualizing. As the score of zero was not independent from mortality (score of zero was the defined humane endpoint and necessitated euthanasia), it was excluded from statistical analyses. Comparing the categorical scores at 18 and 24 HPC indicated a significant relationship between score and survival (Fisher's exact test, $p < 0.001$). A simple linear regression model examining percent overall survival against assigned score at 24 HPC (mice with a score of 0 were excluded) was found to be significant ($R^2 = 0.946$, $p = 0.0035$, $n = 210$) with each increase in

very good R square, why use other machine learning? what's the final observation time point of death ?

score equating to a $22\% \pm 8\%$ improved chance of survival (Fig 2A). Visualizing the changes in score over the first 48 hours of disease (capturing 93% of deaths) showed a clear separation between survivors and non-survivors beginning around 18 HPC, with higher scores associated with recovery and survival (Fig 2B).

Figure 2. Health scores are directly associated with outcome in a polymicrobial model of sepsis in neonatal mice.

(A) Scores recorded at 24 hours post challenge (HPC) plotted against the percent of mice which survived the IP cecal slurry challenge. Scores of zero were excluded as they were the pre-determined humane endpoint. A significant linear regression was found ($R^2 = 0.946$, $p = 0.0035$, $n = 210$) with percent survival increasing by $22 \pm 8\%$ for each single point increase in score. (B) Visual representation of change in score over time following IP cecal slurry challenge.

The inability to distinguish survivors through health scores at 12 HPC (Fig 2B) was consistent with qualitative observations that most mice appeared similar during this time period irrespective of final outcome. At 24 HPC, the scores were well distributed with the least frequent score (3) assigned to 10.9% of mice (29/266) and the most frequent, the healthiest score of 5, assigned to 21.0% (56/266) of mice (Fig S1). At 18 HPC, many of the mice appeared similarly ill, with only 6.9% at the healthiest score of 5 and 46.8% at the middle score of 2. Thus, health scores at 24 HPC are most useful in their ability to capture a range of sickness in neonatal mice.

Figure S1. Frequency of scores observed at 18 and 24 hours post challenge.

Assigned health score distribution 18 and 24 hours after IP cecal slurry challenge of neonatal mice. Disease progression towards resolution or humane endpoint was demonstrated by the relative increase in scores on either extreme (0 and 5).

To test the validity of our scoring system, a separate cohort of mice generated under the same experimental conditions were sacrificed at 24 HPC and samples of whole-blood, spleen, liver, and lung tissue were plated on sheep's blood agar and the colony forming units (CFU) growth was assessed 24 hours later. Two separate, one-way multivariate analyses of variance (MANOVAs) were performed to assess the relationship between categorical scores at 18 and 24 HPC and bacterial load across all four biological samples. A significant MANOVA effect was attained both at 18 HPC (Pillais' Trace = 0.24, $F(20, 572) = 1.85$, Bonferroni-adjusted p-value = 0.0276) and 24 HPC (Pillais' Trace = 0.44, $F(20, 5.72) = 3.50$, Bonferroni-adjusted p-value < 0.001) (Fig 3). The distribution of scores broadened as disease progressed towards an outcome – nearly all mice scored between 1-3 at 12 HP, most fell within that same range at 18 HPC, but an even distribution was present at 24 HPC (Fig S1). Our health scores were strongly associated with both survival and bacterial load at 24 HPC.

Figure 3. Health scores at 18 and 24 HPC are related to bacterial load at 24 HPC.

(A) A comparison of score 18 hours post challenge (HPC) with bacterial load in whole-blood and tissues showed a significant MANOVA effect (Pillais' Trace = 0.24, $F(20, 572) = 1.85$, Bonferroni-adjusted p-value = 0.0276) indicating score at 18 HPC is related to the bacterial load measured in tissues of the same animals collected 6 hours later (24 HPC). (B) A significant MANOVA effect was also attained at 24 HPC (Pillais' Trace = 0.44, $F(20, 5.72) = 3.50$, Bonferroni-adjusted p-value < 0.001) demonstrating the relationship between health scores and bacterial load across all biological compartments.

220 **Righting reflex** and **mobility** are **predictive** of outcome **prior to**
 221 **sacrifice**

222 We used a **machine learning approach** to **classify** mice as survivors or non-survivors **based primarily on**
 223 **righting reflex**, **mobility**, and **weight**, specifically looking at how each metric **changed** **from 18 to 24 HPC**.
 224 Only 12% (28 / 222) of mice were *at* humane endpoint at 24 HPC, so while the association between
 225 score components and outcome may be influenced by the inclusion of these mice (mice which were FTR
 226 non-mobile on both sides are at humane endpoint and **must be sacrificed**) they did **not** represent a large
 227 enough proportion to call the results into question.

Where is feature?

229 We used **10-fold cross validation** training on a combination of six algorithms and three methods of
 230 **feature selection** in order to identify the **most effective** method of classification (Table S1). A more
 231 detailed description of this approach is available in the **supplemental section**. For the final predictions,

Where is well?

232 we used **Gradient Boosting Machine** as our classifier, combined with **features selected by Pearson**
 233 **correlation coefficients**. This approach had a cross validation average **accuracy** of 0.91 with a low
 234 standard deviation of 0.06, indicating little risk of over or underfitting (Fig S2). With the

F1

235 hyperparameters selected for our Gradient Boosting model, we built a model on the **training data subset**
 236 with known outcome and applied it to 33% of the original data set **not** used during the **training phase**.

237 With a total of 74 cases to classify as “survivor” or “non-survivor”, the **Gradient Boosting model**
 238 performed well with an **accuracy** score of **0.85** and an **AUC** of 0.93, with 33 true positive cases classified
 239 as “non-survivor” and 30 cases classified as “survivor” (Fig 4). To further **validate** our **model**, we
 240 generated an **additional dataset** with the same data collection process as well as the same data cleaning
 241 and standardization pipeline. With a total of **21** new data **points**, our model was able to accurately
 242 classify 18 with an average score of **85% accuracy** (Fig S3). **Righting reflex** and **mobility** at 24 hours

together with **change in weight** were the features most strongly correlated with outcome (Fig S4).

Finally, we applied our classifier to the subset of mice which were sacrificed at 24 HPC and identified a clear **relationship** between **survival** and **bacterial load** (Fig 5).



Figure 4. Righting reflex, mobility, and biometric data can predict **survival** or **non-survival** prior to sacrifice in neonatal mice.

(A) Confusion matrix showing the accuracy of the classification model applied to the test dataset, the model is accurately able to distinguish the survivor class from the **non-survivor** class with an 85% accuracy. (B) **ROC curve** illustrating the sensitivity of the model applied to the test set.

Figure 5. Bacterial load is strongly associated with predicted outcome.

only related with bacterial load or damage/operation stress?

Neonatal mice injected IP with cecal slurry were sacrificed **24 hours** post challenge, with tissue and blood samples plated on **sheep's blood agar** at time of sacrifice. Mice were predicted to **survive** or **not survive** based on **righting reflex**, **mobility**, and other **biometric data**. Mice which were predicted to **survive** had a significantly lower bacterial load in all measured biological compartments than predicted non-survivors (Independent two-group t-tests with Bonferroni correction, $p < 0.001$). **All** mice with **zero bacterial load** in the whole-blood (9) were **predicted** to **survive**.

Discussion

Here we presented a system of **health scores** for neonatal mice which were used to confidently predict mortality 24 HPC in an established model of neonatal **sepsis** [15]. The **significance** of this is **twofold**: 1) the classifier enables investigators to include "clinical" outcome (e.g. mortality) as a covariate of interest

when all animals are sacrificed, and 2) the scoring system can represent a new standard in recording and reporting neonatal mouse health which has implications beyond this specific model.

We also demonstrated that survival can still be assessed in neonatal mice which were all sacrificed 24 HPC; without such an approach, one is forced to rely on biomarkers (i.e. inflammatory cytokines, cell mobilization, etc.) as a proxy outcome. Given that it is increasingly unclear whether death from sepsis is associated with an inflammatory or an anti-inflammatory response [2,16–19], it is critical to minimize reliance on biomarkers without some rigorous evidence of their relevance to survival. Our choice of bacterial load as an outcome of interest served a dual purpose, it: 1) validated the legitimacy of the scoring system and classifier, as a correlation between outcome and bacterial load is often assumed [9,20], and 2) provided evidence that there truly is a link between bacterial load and outcome in this model of polymicrobial sepsis. Thus, we have removed one assumption which underlies neonatal mouse work. While a link between outcome and bacterial load may seem unsurprising, this demonstrates a proof of concept for situations wherein one must rely on a less obvious biomarker as a proxy outcome (i.e. sterile inflammatory models, inactivated bacteria, etc.). These health scores and classification approach provide a direct mechanism for assigning survival and non-survival rather than solely observing inflammatory markers and assuming a relationship which may or may not exist [14].

We were surprised to see that righting reflex in of itself was strongly correlated with bacterial load across all measured biological compartments, indicating it was a robust metric of neonatal mouse health (Fig 1). Given the simplicity of collecting, these data suggest it should be recorded whenever neonatal mouse well-being is in question. Righting reflex in neonatal mice has previously been used as a tool to study behavioral development but has never been used to measure disease progression nor assess overall health in the context of an infection [21]. These data represent an important contribution

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towards describing neonatal mouse health which can help to inform ethical decisions around humane endpoints.

A potential limitation of this study is that mice which were non-mobile and failed to right on both sides (score 0) were sacrificed for ethical concerns, potentially violating the assumption of independence between score and survival. We addressed this by excluding mice at score 0 from statistical analyses when survival was the outcome of interest. Further, the high bacterial load observed in whole-blood and the various organs independently validated the biological relevance of the scoring system presented. The use of righting reflex as a decision point in humane endpoint can help to standardize mouse models and reduce unnecessary suffering for newborn mice. In fact, the extremely low survival rate (<10%) of mice which failed to right and were lethargic 24 HPC suggests that this should be used as the humane endpoint for future studies. There is also some question surrounding the use of righting reflex as a metric in younger mice which may be unable to right even at a healthy state – further research is warranted prior to assigning these health scores to mice prior to DOL 7.

The availability of a robust scoring system, with the additional proof that it can be predictive of outcome, should allow for a stratification of experimental groups beyond treated / untreated, or challenged / unchallenged, but also treated survivors / treated non-survivors, etc. The latter comparison will provide mechanistic insight into responders and non-responders of novel treatments – precisely what one hopes to attain from working with an animal model. Here we demonstrated that bacterial load across all measured biological compartments was strongly correlated with our predictive health scores and therefore final outcome in neonatal, polymicrobial sepsis. This study represents a blueprint for how to report neonatal mouse health in infectious disease models, and how to use that data to examine survival in mice which had to be sacrificed for sample collection.

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Supporting Information

Figure S1. Distribution of scores assigned to neonatal mice at 18 and 24 hours post challenge.

Scores assigned to neonatal mice 18 and 24 hours post IP challenge with cecal slurry (HPC). At 18 HPC the scores are poorly distributed, with the vast majority of mice assigned a score of 2 (failure to right, mobile) indicating that most mice have not progressed towards survival or non-survival. By 24 HPC the scores are evenly distributed as mice have begun to succumb to or recover from sepsis.

not statistic words

Table S1. Accuracy of different algorithms and feature selection approaches in classifying mice into survivors or non-survivors at 24 HPC.

feature or parameter?

Method	GridSearch (No feature selection)	Correlation + GridSearch	Hypothesis + GridSearch	Model based + GridSearch
Logistic Regression	0.860	0.874	0.874	0.874
K Nearest Neighbor	0.827	0.833	0.851	0.856
Decision Tree	0.820	0.842	0.842	0.869
Random Forest	0.878	0.869	0.874	0.878
Gradient Boosting	0.883	0.887	0.887	0.883
XGBoost	0.883	0.892	0.892	0.887
Average	0.859	0.866	0.870	0.875

Figure S2. 10-fold cross validation of Gradient Boosting Machine algorithm.

Figure S3. External validation of Gradient Boosting Machine algorithm.

(A) Confusion matrix demonstrating the predictive capacity of the classifier when applied to a never seen dataset of 21 neonatal mice challenged with the cecal slurry model of sepsis. The classifier accurately categorized 9/11 survivors and 9/10 non-survivors, in line with the internal cross-validation

set. (B) Confidence of predicted outcome, survivor or non-survivor, within the 21-mouse external validation set.

Figure S4. Feature selection visualization using Pearson correlation.

Heatmap of feature correlations of with Pearson correlation. Scores were split into components, starting with looking at righting reflex and mobility independent from one another and then further separated by the lower and higher measurements of each score taken in duplicate. Change in righting reflex reflects the difference between the monitoring timepoints at 18 HPC and 24 HPC.

Figure S5. Distribution of training accuracy per classifier family

Classifier construction

We collected the experimental data per mouse and mapped it to the outcome of the results at 24 HPC.

Outcome categories were defined as survivors and non-survivors. The raw data collected was cleaned

and preprocessed before building the model. Attributes that were not useful for classification, such as

unique identifiers, dates and times, were removed. The final dataset was well-balanced, containing 112

non-survivors and 110 survivors. Following feature cleaning and engineering (i.e. separating scores into

categorical components) the dataset had 23 attributes. We tested and compared six different algorithms

which fit within two distinct approaches; baseline learning algorithms (Logistic Regression, K Nearest

Neighbors and Decision Tree) and ensemble learning algorithms (Random Forest, Gradient Boosting and

XGBoost). Feature independence was evaluated and visualized using Pearson correlation coefficients

(Fig S4).

real world is not balanced

Except few attributes highly correlated, we can observe a certain attributes independence, which could

be beneficial for the final accuracy of the trained model. Highly correlated features may be a bias in the

training phase, as they describe the data point the same way and bring no variability to learn different

patterns which may cause either overfitting or underfitting the model to the training set. We tested

three methods of feature selection (Pearson correlation, K nearest neighbor, support vector

classification) and Grid Search optimization. We performed 10-fold cross validation training using all

mentioned items measured the training accuracy for each use case (Table S1).

The accuracy was high for all the algorithms used (>0.8) which indicates the prediction of outcome to

be a solvable problem. Gradient Boosting and XGBoost with Pearson correlation as feature selection

where is the code? which languages/software?

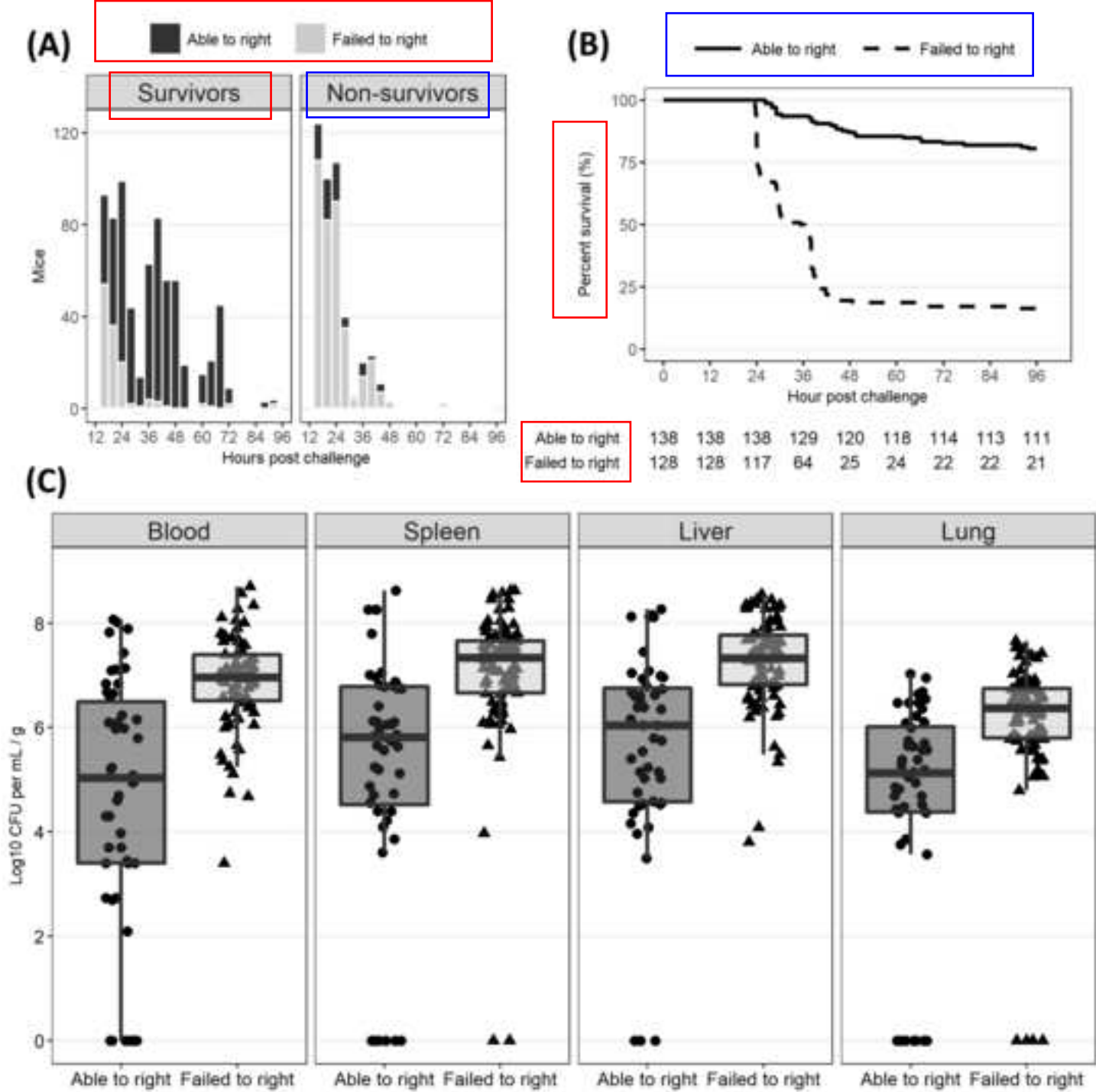
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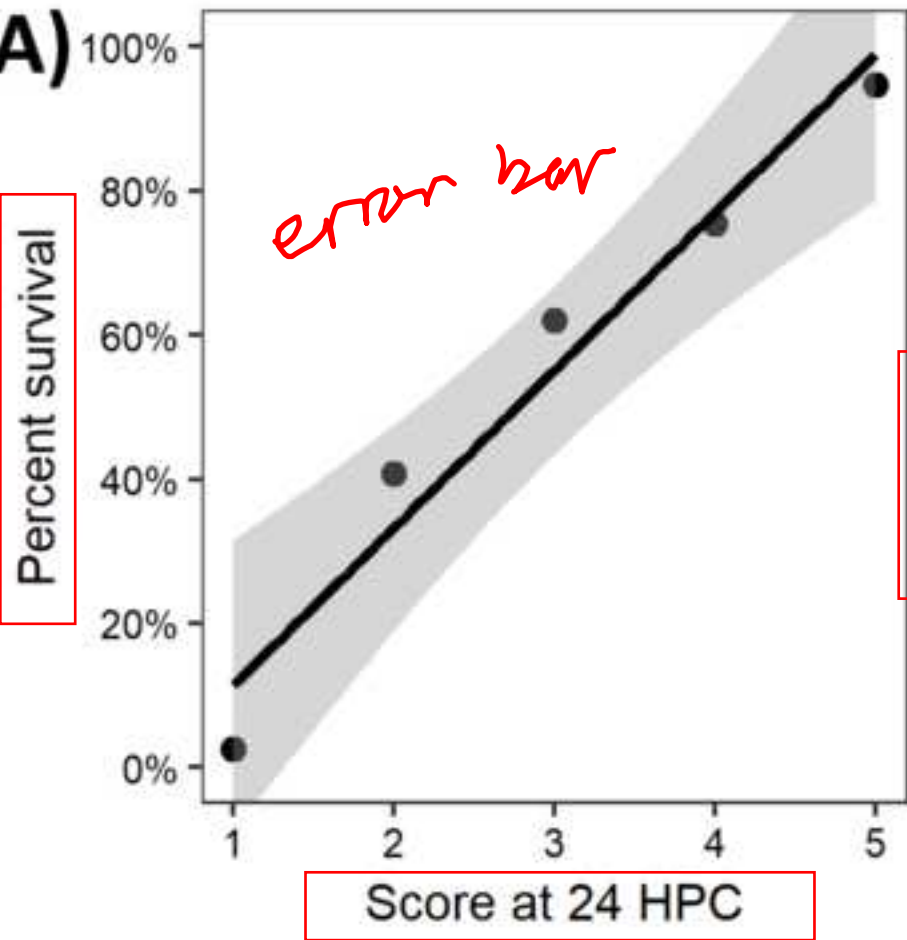
gave the best training accuracy compared to the other methods. We thus decided to select Correlation and Gradient Boosting (Fig S5).

There was a convergence to a cross validation average accuracy of 0.91 with a low standard deviation of 0.06, indicating high accuracy with little risk of over or underfitting. With the hyperparameters selected for our Gradient Boosting model, we built a model on the training dataset and applied it to a 33% of the original data set not used during the training phase. With a total of 74 cases to classify as “survivor” or “non-survivor”, Gradient Boosting model performed well with an accuracy score of 0.85 and an AUROC value of 0.93, with 33 True Positive cases classified as “non-survivor” and 30 cases classified as “survivor”. The model was further validated with a truly external set of 21 mice wherein the previous accuracy was again seen (0.86) (Fig S3).

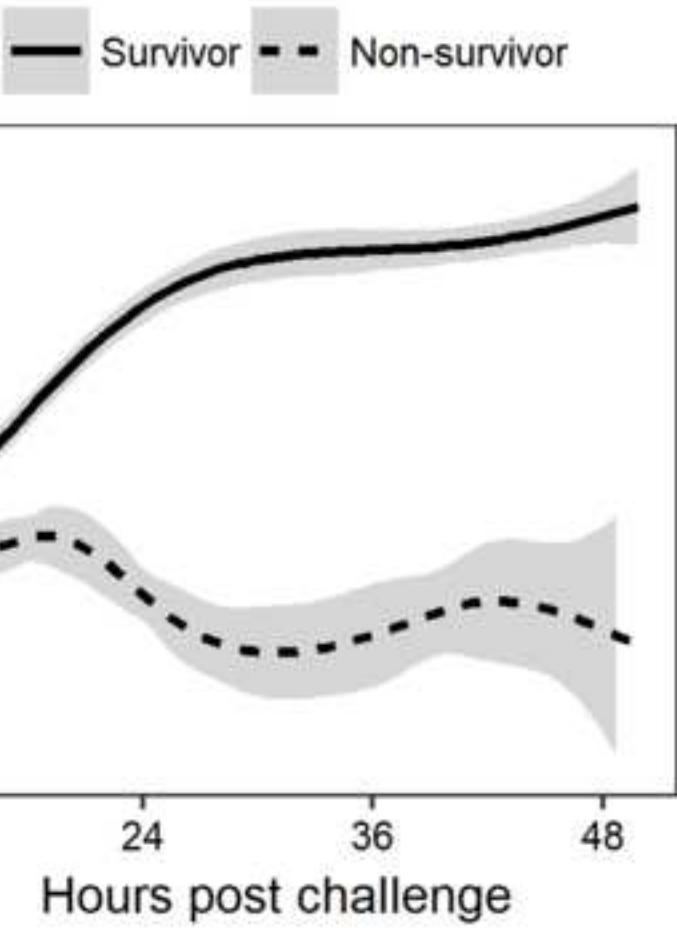
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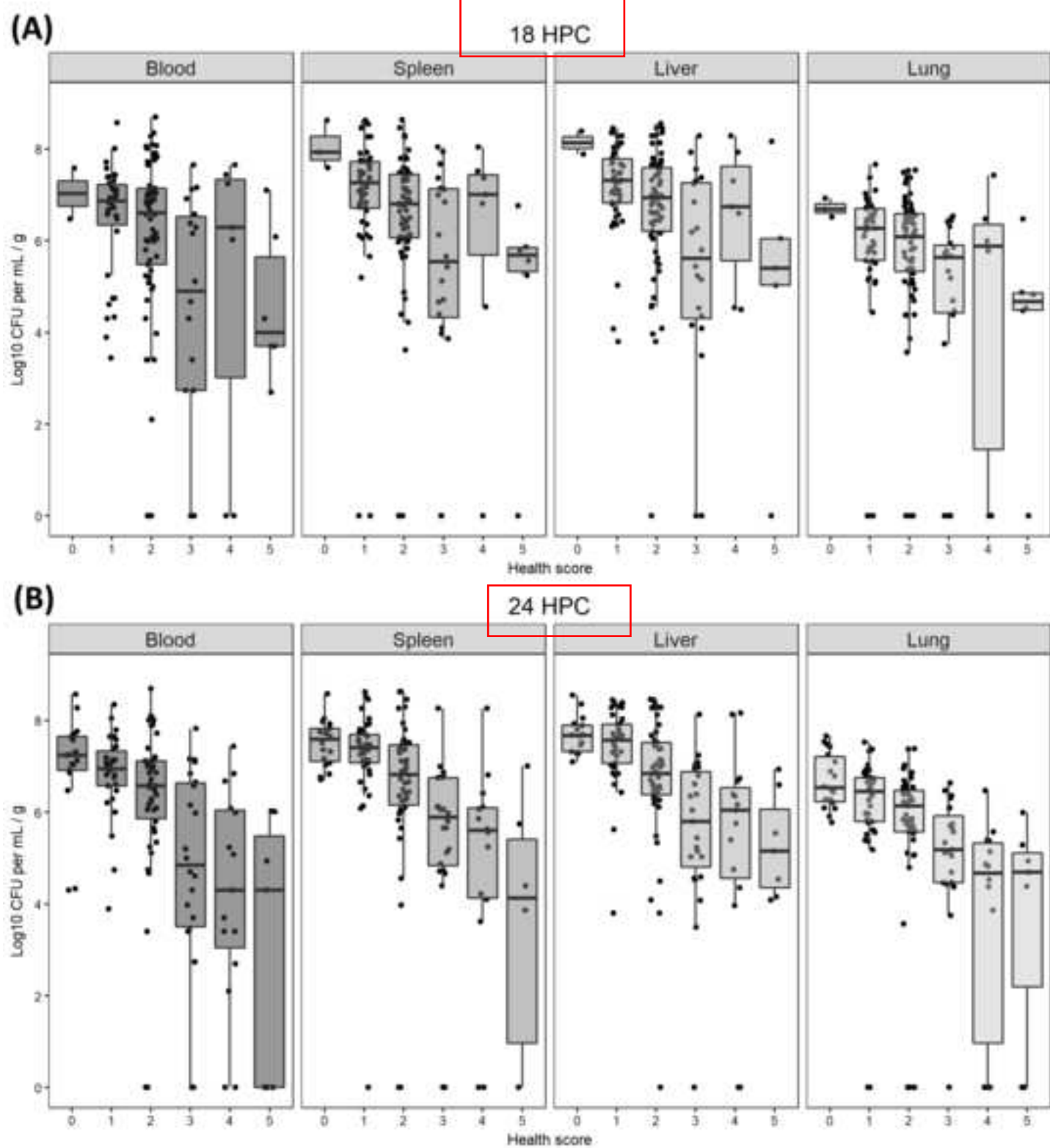


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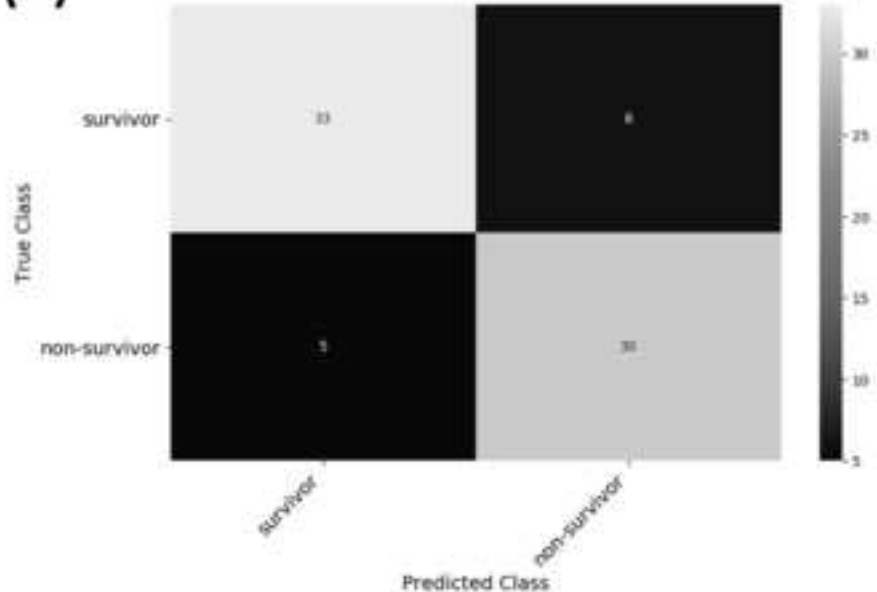


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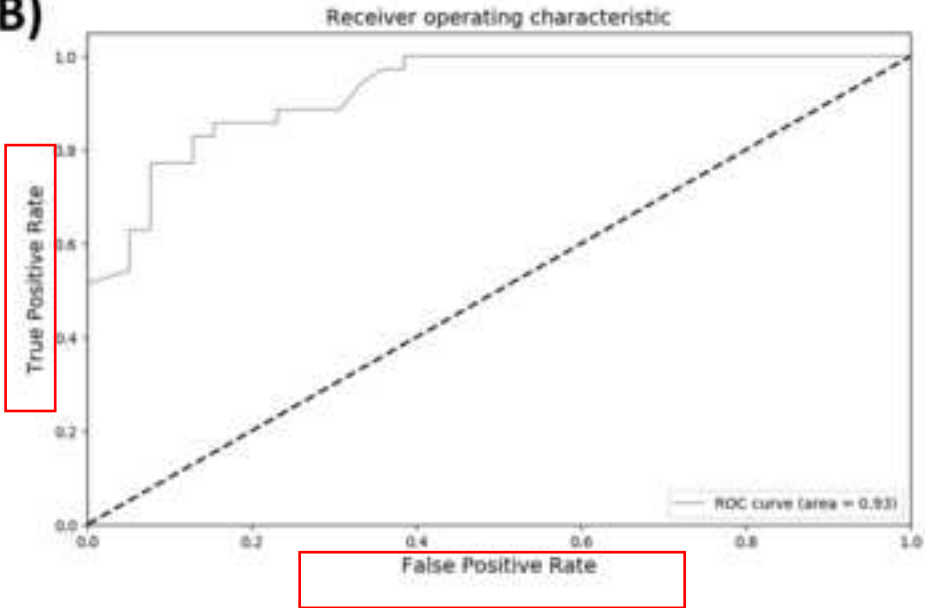




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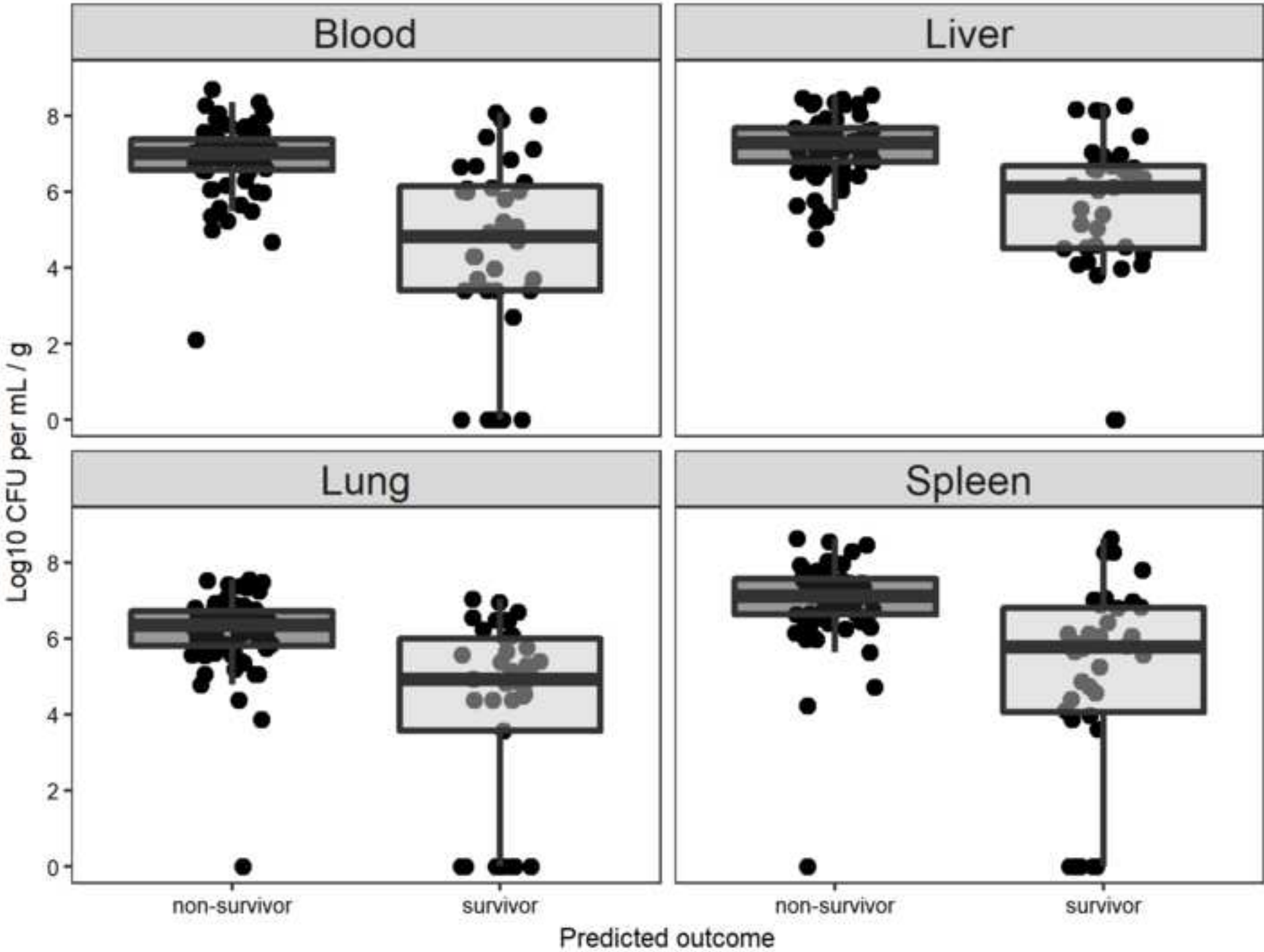


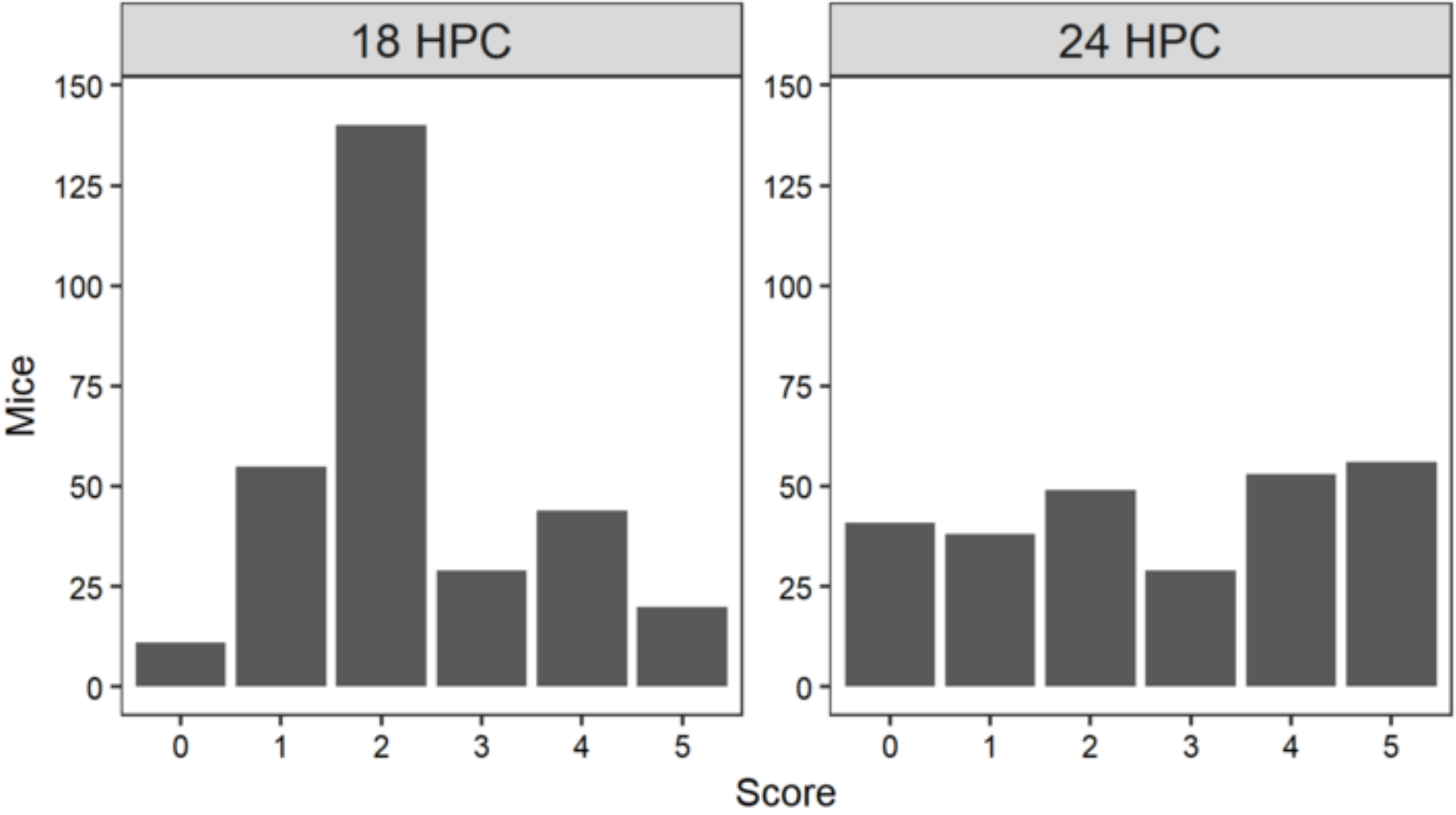
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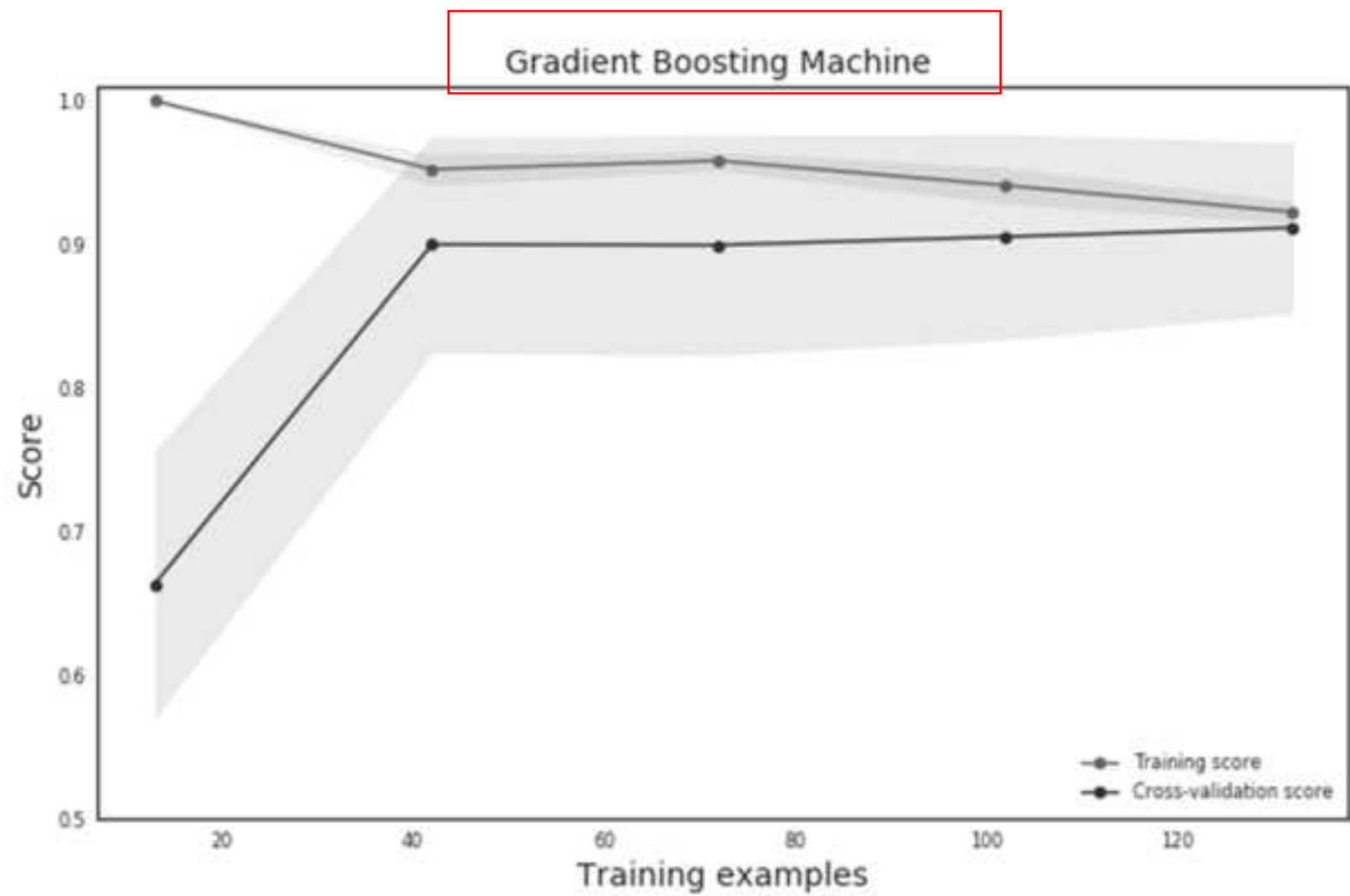


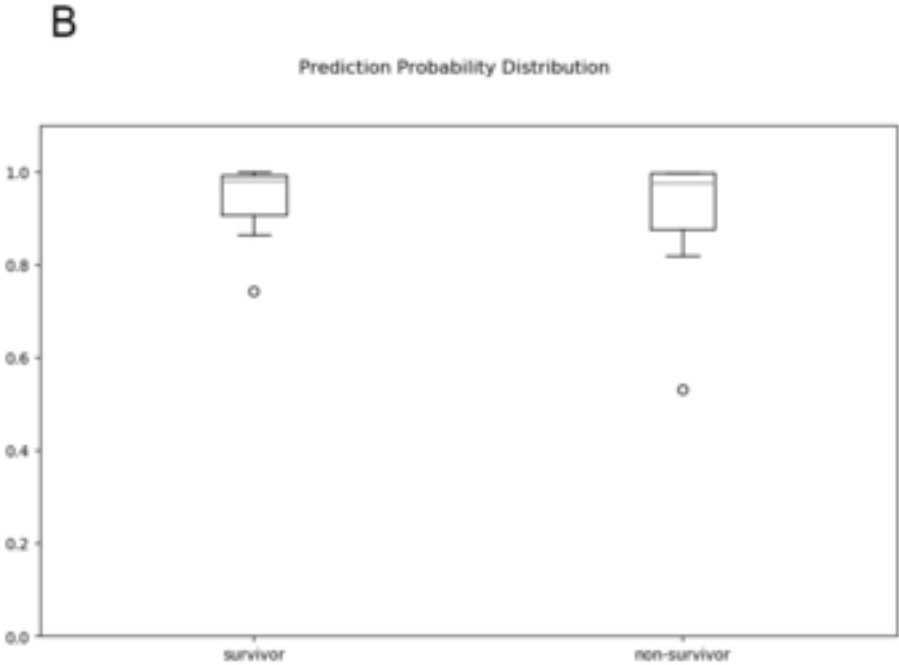
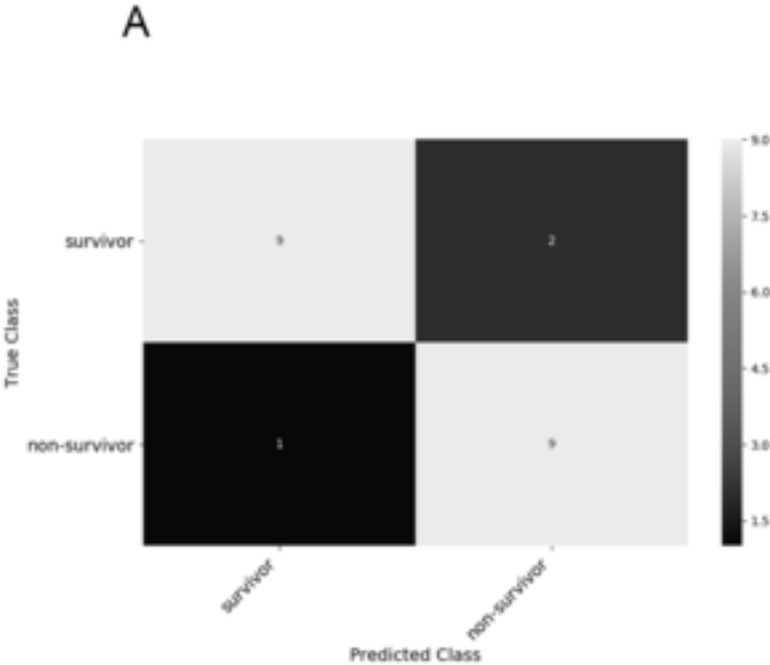
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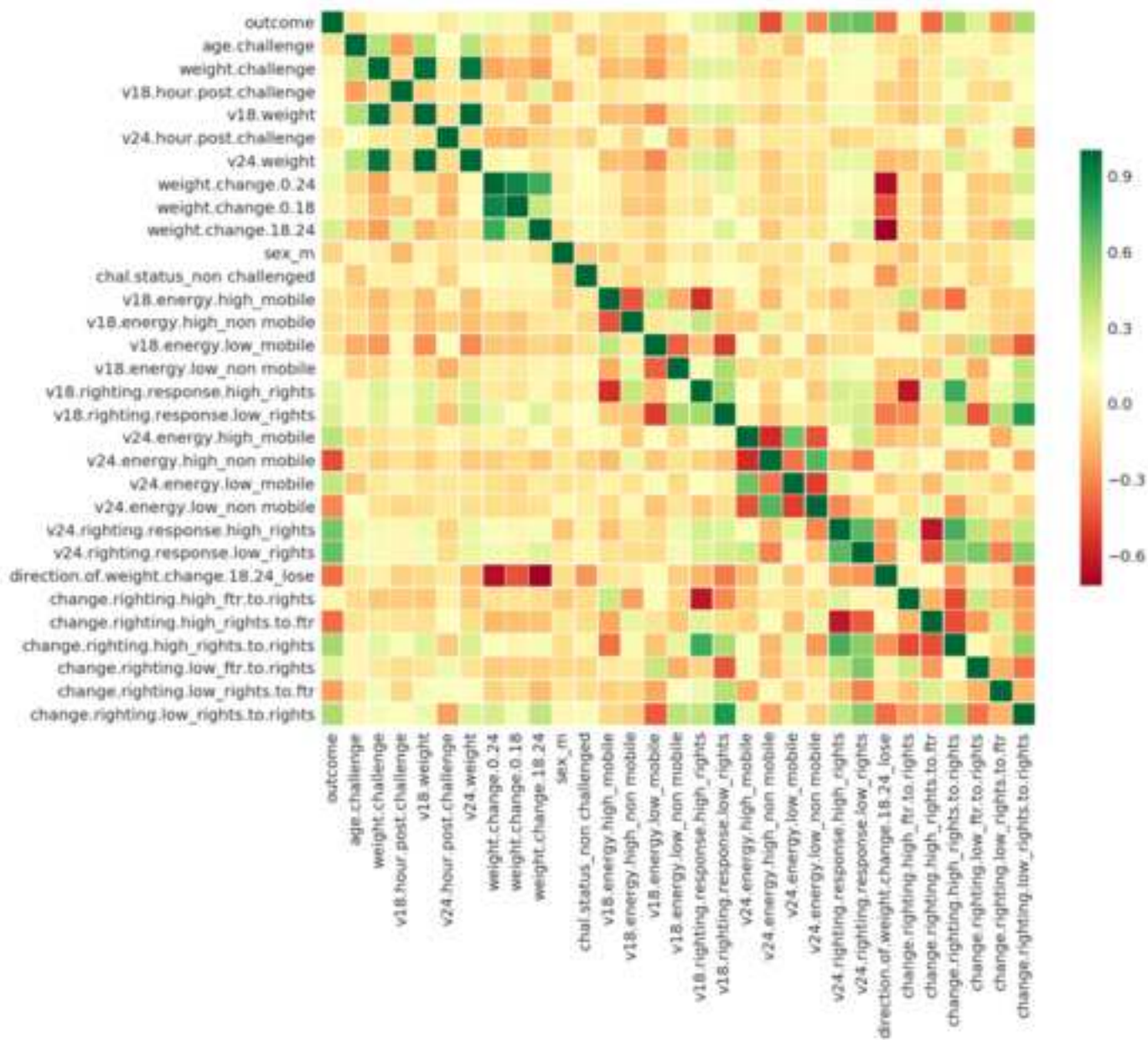








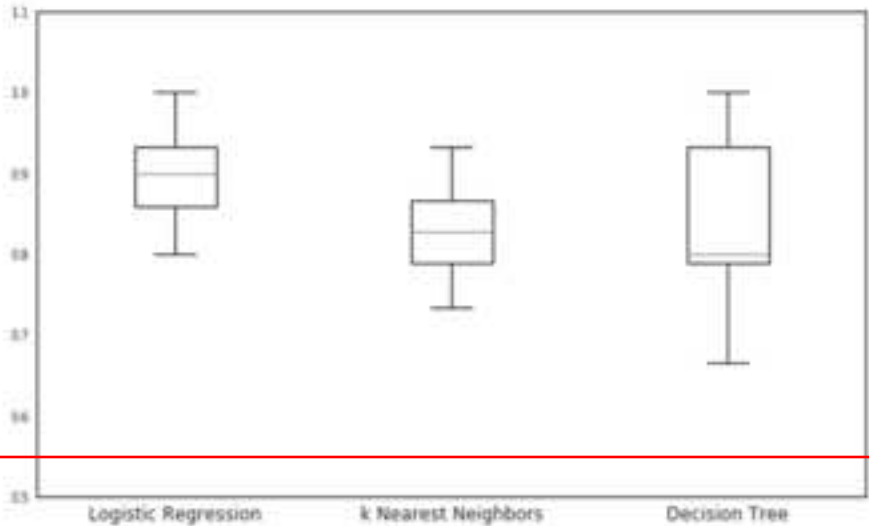
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What?

A

Baseline Algorithms Comparison



B

Ensemble Algorithms Comparison







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